ABSTRACT OF THE DISCLOSURE

A method and system for encoding an analog signal directly into a sequence of high-cardinality data words for transmission or storage. The system bypasses the traditional step of analog-to-digital conversion (ADC) into a binary format. Instead, samples of an analog waveform are directly mapped to discrete states in a high-radix, parallel data transmission system, such as the wavelength-encoded states of a parallel optical interface. Each state, represented by a pixel with a specific color or wavelength, corresponds to a specific amplitude or other characteristic of the analog signal at a sampling point. This method allows for a more efficient representation of analog data, enabling extremely high-density storage and transmission, potentially allowing a short song to be stored in a few frames of optical data.